Robert Orlikowski

Software Engineer

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Technical Skills

Languages C, C++, Python
Tools Git, GitLab CI, Docker, CMake, Conan, Meson, GDB

Experience

Dolby Laboratories

Associate Engineer **April 2025 – June 2026**

- Implemented multiple features and extended functionality for C++ command-line audio processing tools used in professional encoding workflows.
- Maintained and improved loudness processing libraries, including CI pipelines, packaging infrastructure, and Conan package distribution.
- Refactored a loudness processing library to improve maintainability, simplify implementation, and reduce complexity.

Software Engineering Intern **March 2024 – April 2025**

- Maintained and extended a Windows based C++ command-line tool used for interoperability testing.
- Ported tooling from Python reference implementations to C++ applications.
- Added PyTorch backend support to a signal-processing library and redesigned backend integration to simplify future extensions.
- Designed and implemented cross-platform build and testing pipelines.

Projects

AliceDB | C++, STL, io_uring, CMake, GTest

- Incremental in-process streaming database for C++ that supports relational transformations over change streams.
- Designed worker pool for parallel processing of independent graphs and nodes.
- Implemented relational operations including joins, aggregation, filtering, projection, set operations, and distinct queries.
- Wrote persistent storage layer with buffer pool management using io_uring for asynchronous I/O.

Inference Engine | Python, PyTorch, Triton, ZeroMQ, FastAPI

- LLM inference serving engine supporting CPU and CUDA backends.
- Implemented continuous batching, chunked prefill, and prefix KV cache optimization.
- Added custom Triton based kernels and CUDA graph replay for faster inference on GPUs (300+ tokens/s on RTX 5070 for Qwen 2.5 0.5B)
- Designed multiprocess architecture with ZeroMQ based communication and OpenAI-compatible API endpoints.

rpiOS | C, ARMv8-A, Raspberry Pi 3B+, QEMU, GDB

- Operating system primitives targeting the ARMv8-A architecture on Raspberry Pi 3B+.
- Implemented virtual memory management, interrupt handling, process scheduling, and read-only filesystem support.

Education

University of Wrocław **2020 – 2024**

BSc in Individual Studies in Mathematics and Computer Science.

Mathematics Key courses:

- Real Analysis I, II, III, Probability, Statistics, Stochastic Modeling, Numerical Analysis, Differential Equations, Abstract Algebra, Linear Algebra, Discrete Mathematics.

Computer Science Key courses:

- Compiler Construction, Operating Systems, Computer Architecture, Linux Device Drivers, Computer Networks, Neural Networks and Natural Language Processing, Machine Learning, Databases, Algorithms and Data Structures, Programming Methodologies, Logic.